Chittagong University of Engineering & Technology

Department of Electronics and Telecommunication Engineering

EXPERIMENT NO: 03

Name of the Experiment

Configuration of Routers as Default Gateway in LAN

Course No : ETE 452

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Objectives

- To configure a router for inter-network communication.
- To understand the role of default gateway in a LAN.

Software

- Cisco Packet Tracer

Introduction

A router is used to connect multiple networks together.The default gateway allows devices to communicate with external networks.Routing enables packet forwarding between different IP networks.

Procedure

1. Multiple PCs, a switch, and a router were placed in the workspace.
2. PCs were connected to the switch using straight-through cables.
3. The switch was connected to the router interface.
4. IP addresses were assigned to PCs and router interfaces.
5. The router IP was set as the default gateway for all PCs.
6. Connectivity was tested using ping across networks.

Connection Diagram

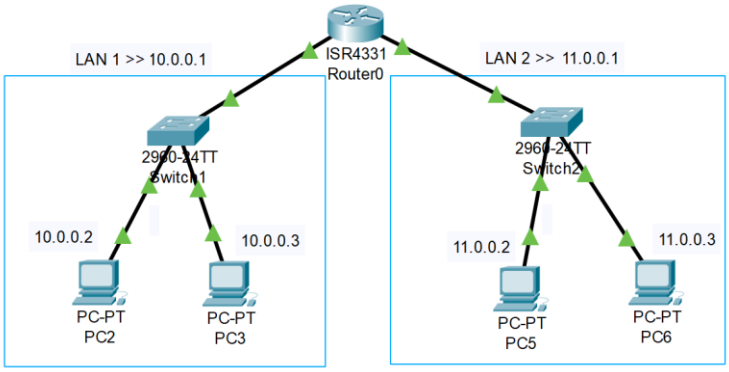

Figure 3 : Local Area Network (LAN) using (a) Hub and (b) Switch

Discussion

1. Successful communication between different networks was achieved using a router.
2. The importance of default gateway configuration was clearly observed.
3. Without a default gateway, inter-network communication was not possible.
4. Router interfaces were required to be properly configured and enabled.
5. This experiment demonstrated the basic concept of routing in a LAN environment.